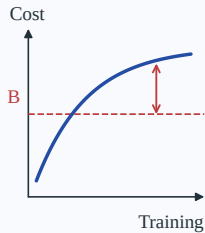


Soft penalty

costs in the objective

$$J_r - \sum_k \lambda_k J_{c_k}$$



Budget leakage

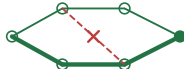
schematic

SafeOT-Dual

K constraints · one joint solve



**ONE joint
flow solve**



$$\lambda_k = \lambda_k^* + \beta_k$$

joint prices

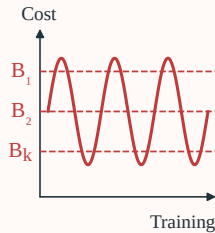
cost feedback

Actor update

Explicit budgets

coupled constraints

$$J_{c_k} \leq B_k$$



Budget oscillation

schematic

Fixed penalty

$c = 0$; fixed β

One price rule: exact settings and conditional limits
Lagrangian

$c = 0$; adaptive β

SafeOT-Dual

finite ϵ

LP / active-set

$\epsilon \rightarrow 0$ / saturated clip
under assumptions